Bugada Roopansha

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.07
Sri Chaitanya Junior Kalasala	Class XII (TSBIE)	2024	95.70%
Sri Chaitanya Techno School	Class X (SSC)	2022	9.70%

ACHIEVEMENTS

Represented **IIT Hyderabad** at **e-YRC** 2025-2026 InterNational Robotics Competition and secured **3** rd place.

SKILLS

Programming Languages : C, Python, PostgreSQL and Latex

Libraries : Numpy, Pandas, Matplotlib, OpenCV and PyTorch

Developer Tools & Frameworks: Git, Linux and ROS 2

Soft Skills: Team Player, Communication, Collaboration and Problem-Solving

RELEVANT COURSES

- **Artificial Intelligence**: Foundations of Machine Learning, Convex Optimization-I, Convex Optimization-II, Deep Learning and AI For Humanity.
- **Mathematics**: Probability and Random Variables, Vector Calculus, Differential Equations, Calculus-I, Calculus-II and Series Of Functions.
- **Computer Science**: Data Structures, Algorithms and Database Management system.
- **Electrical Engineering**: Control systems, Linear Systems & Signal Processing and Matrix Theory.

PROJECTS

Holo Battalion | e-Yantra Robotics Competition, IIT Bombay :

- Developed a ROS 2-based multi-robot warehouse automation system where three holonomic robots autonomously sorted and transported color-coded crates using overhead ArUco marker tracking.
- Implemented **A*** path planning algorithm for autonomous robot navigation and task coordination across both simulation and hardware stages.
- Engineered an overhead camera-based localization pipeline using **ArUco** marker detection (**OpenCV**) for real-time positional feedback, enabling precise closed-loop control of each bot on the arena.

Industrial AI Surface Inspection | PyTorch, OpenCV, Computer Vision

- Built an end-to-end computer vision system for automated industrial surface defect detection using **PyTorch**, **OpenCV**, and transfer learning on real-world manufacturing datasets.
- Designed the complete training pipeline including **data preprocessing, augmentation, fine-tuning of pretrained CNN architectures, performance evaluation, and Grad-CAM explainability** for interpretable defect classification.
- Developed a production-style inference application capable of processing unseen images, generating confidence scores, visualizing detected defects, and supporting scalable AI-assisted quality inspection.

Experimental Hub | FastAPI, PostgreSQL, MCP

- Built a full-stack AI experiment management platform for organizing machine learning experiments, tracking metrics, managing datasets, and versioning trained models through a scalable web application.
- Designed and implemented a **FastAPI** backend with **PostgreSQL**, providing RESTful APIs for experiment lifecycle management, artifact storage, and performance analysis.
- Integrated an **MCP-powered AI assistant** capable of understanding natural language queries to search experiments, summarize training results, compare model performance, and streamline AI development workflows.

Foundations of Machine Learning (Coursework) :

- Implemented MLE/MAP estimation, polynomial regression, and sequential Bayesian inference with posterior predictive uncertainty, studying overfitting and bias-variance tradeoff for MNIST dataset.
- Built generative classifiers (GNB, LDA, L2-LDA) and unsupervised models (PCA, K-means++, EM-based GMM) for face reconstruction and clustering, evaluating robustness to noise and class imbalance.
- Built a from-scratch backpropagation/autograd engine (Value class with computation graph construction and gradient flow), along with modular NN components (Linear, ReLU, Sigmoid, Sequential), loss functions, and an SGD optimizer
- Built a mini machine learning library similar to scikit-learn, with tools for loading data, preprocessing, splitting data for testing, and classical models like Logistic Regression and LDA

POSITIONS OF RESPONSIBILITY

Robotix IITH :

- Core Member, Programming & Machine Learning Domain, Robotix IITH.
- Represented IIT Hyderabad at the **e-Yantra Robotics Competition** (2025-26).

Class Representative:

- Serving as the Class Representative for the 2028 batch of the Artificial Intelligence Department.

Teaching Assistant [Matrix theory], IIT Hyderabad :

- Guided students in implementing numerical linear algebra techniques.
- Helped students debug mathematical and programming problems.

Teaching Assistant [Intoduction to programming], IIT Hyderabad:

- Assisted students in learning programming fundamentals using C language.
- Guided students with problem-solving approaches, debugging techniques and coding assignments.